Baharul Islam

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SUMMARY

AI/ML engineer and researcher with experience building production-oriented multi-agent AI systems, RAG pipelines, and security methods for large language models. Skilled in Python, PyTorch, LangGraph, Transformers, retrieval systems, model evaluation and FastAPI, with published research in NLP and computer vision.

EXPERIENCE

Trainee - AI/ML | ITCart Pvt. Ltd. *Apr 2026 – Present*

- Working on development of a multi-tenant and human-governed HR decision platform using **LangGraph** for multi-agent orchestration, **LangSmith** for observability and evaluation, tenant-scoped RAG, deterministic policy controls, and human-in-the-loop workflows.
- Working on development of a Learning Management System using **FastAPI, Qwen3, Haystack, Qdrant, and SQLite**, supporting PDF ingestion, parsing, chunking, embeddings, course and quiz generation, grounded tutoring, quiz grading, feedback, and learner analytics.
- Implemented tracked multi-step APIs, background job processing, hybrid dense and BM25 retrieval with reciprocal-rank fusion, schema-validated LLM responses, failure fallbacks, automated tests, Docker packaging, and Azure-ready deployment configuration.

Research Intern | The University of Adelaide | Supervisor: Dr. Srinibas Swain *Jan 2026 – Jun 2026*

- Worked on **backdoor detection and removal methods for large language models** in realistic deployment settings, where models behave normally on clean inputs but produce harmful outputs in the presence of hidden triggers.
- Developed a **self-guided backdoor detection and removal framework** for LoRA-adapted Transformer classification and generation models using only unlabeled test data.
- Localized suspicious LoRA layers and modules by measuring changes in attack success and signature rates, then performed targeted repair using **knowledge distillation, backdoor suppression, and stability regularization** while preserving clean model performance.

Research Intern | Khalifa University | Supervisor: Dr. Muhammad Owais *Sep 2024 – Jan 2025*

- Implemented and evaluated LLM applications for VR therapy, marine conservation, disaster management, and other underexplored domains.
- Conducted systematic literature reviews and identified technical, ethical, adoption, and deployment challenges across interdisciplinary use cases.
- Proposed practical solution frameworks, deployment strategies, and future research directions for domain-specific foundation models.

SKILLS

Programming and Data: Python, Java, C, SQL, NumPy, Pandas, data preprocessing, feature engineering, model evaluation
Machine Learning and Deep Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, Transformers, Hugging Face, BERT, fine-tuning, NLP, computer vision
Agentic AI, LLMs, and RAG: LangChain, LangGraph, LangSmith, Haystack, multi-agent systems, model context protocol, prompt engineering, tool calling, structured outputs, workflow orchestration, Qdrant, vector databases, semantic search, reranking, retrieval evaluation
Backend, Cloud, and MLOps: FastAPI, Flask, REST APIs, Pydantic, Streamlit, MySQL, Docker, AWS, Git, GitHub, experiment tracking, model monitoring

PROJECTS

Multi-Agent Research System GitHub

- Built a LangGraph/LangChain workflow with specialized agents for web search, summarization, fact-checking, and report generation.
- Added conditional routing, confidence-based retries, shared state, and structured outputs for reliable multi-step execution.
- Containerized the system with Docker, deployed it on AWS Lambda, and implemented CircleCI CI/CD for testing and deployment.

Dental AI Assistant GitHub

- Built a supervisor-based multi-agent assistant for appointment booking, cancellation, rescheduling, and clinic information retrieval.
- Implemented Pydantic validation, structured tool calls, workflow state management, and Pandas-based appointment processing.
- Designed natural-language workflows for patients and clinic staff, reducing manual handling of routine scheduling requests.

Biomedical Evidence Retrieval and Label Prediction Pipeline

- Built an end-to-end clinical-trial NLP pipeline for multi-label prediction and supporting-evidence retrieval from long reports.
- Evaluated GPT-3.5 fine-tuning, BioBERT, Llama 2, and ANN-based models across classification and retrieval experiments.
- Achieved 78% accuracy, 79% F1-score, and 81% retrieval performance on validation and test data.

PUBLICATIONS

Submitted: *MAP-LLM: A Multi-LLM Automated Pipeline for Adversarial Attack Prevention.* ACM ASIACCS.
Submitted: *Foundation Models Beyond the Usual: Survey Atlas of Underexplored Domains from Theory to Execution.* Artificial Intelligence Review.
Published: *AquaFusionNet: Deep Feature Fusion and Attention for Underwater Defect Detection.* Earth Systems and Environment, Springer Nature, 2025. Link
Published: *NBF at SemEval-2025 Task 5: Light-Burst Attention for Multilingual Subject Recommendation.* SemEval 2025. Link
Published: *Vision Transformer-MLP Framework for Driver Drowsiness Image Categorization.* IEEE GCON 2025. Link

ACHIEVEMENTS

Secured **2nd position** in the *Agentic AI Hackathon*, organized by *Cognizant AI Lab* in collaboration with *IIIT Guwahati*.
Secured **4th position globally** in the *Anti-BAD Challenge: An Anti-Backdoor Challenge for Post-Trained Large Language Models*, a competition for *IEEE SaTML 2026*. Leaderboard
Secured **3rd position globally** in the Machine Vision Innovation Tournament at the *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, 2024. Certificate of Achievement
Placed **9th globally** in SemEval 2025 Task 5 on LLMs for Subject Indexing.
Awarded **First Prize** in the Blind Coding competition organized by the Programming Club at IIIT Guwahati.
Achieved an All India Rank of **11,771** in JEE Advanced and **16,571** in JEE Mains (2022).

EDUCATION

Bachelor of Technology in Computer Science and Engineering, IIIT Guwahati Nov 2022 – Jun 2026
CGPA: 9.06/10
Higher Secondary, Kendriya Vidyalaya Barpeta Apr 2020 – May 2022
Grade: 94.4%